

Thus, either

$$\sin x = 0 \quad \text{or} \quad 2 \cos x + 1 = 0;$$

that is, $\sin x = 0 \quad \text{or} \quad \cos x = -\frac{1}{2}.$

The solutions of these equations for $0 \leq x \leq 2\pi$ are

$$0, \quad \pi, \quad 2\pi, \quad 2\pi/3, \quad 4\pi/3.$$

The two solutions $x = 0$ and $x = 2\pi$ tell us that there are horizontal tangent lines at the endpoints of the interval $[0, 2\pi]$. The remaining solutions $2\pi/3$, π , and $4\pi/3$ are the x -coordinates of the points P , Q , and R shown in Figure 3.30. Using $y = \cos 2x + 2 \cos x$, we see that horizontal tangent lines occur at the points

$$(0, 3), \quad (2\pi/3, -1.5), \quad (\pi, -1), \quad (4\pi/3, -1.5), \quad (2\pi, 3).$$

If only approximate solutions are desired, then, to the nearest tenth, we obtain

$$(0, 3), \quad (2.1, -1.5), \quad (3.1, -1), \quad (4.2, -1.5), \quad (6.3, 3).$$

EXERCISES 3.6

Exer. 1–6: Use the chain rule to find $\frac{dy}{dx}$, and express the answer in terms of x .

1 $y = u^2; \quad u = x^3 - 4$

2 $y = \sqrt[3]{u}; \quad u = x^2 + 5x$

3 $y = 1/u; \quad u = \sqrt{3x - 2}$

4 $y = 3u^2 + 2u; \quad u = 4x$

5 $y = \tan 3u; \quad u = x^2$

6 $y = u \sin u; \quad u = x^3$

Exer. 7–62: Find the derivative.

7 $f(x) = (x^2 - 3x + 8)^3$

8 $f(x) = (4x^3 + 2x^2 - x - 3)^2$

9 $g(x) = (8x - 7)^{-5}$

10 $k(x) = (5x^2 - 2x + 1)^{-3}$

11 $f(x) = \frac{x}{(x^2 - 1)^4}$

12 $g(x) = \frac{x^4 - 3x^2 + 1}{(2x + 3)^4}$

13 $f(x) = (8x^3 - 2x^2 + x - 7)^5$

14 $g(w) = (w^4 - 8w^2 + 15)^4$

15 $F(v) = (17v - 5)^{1000}$

16 $s(t) = (4t^5 - 3t^3 + 2t)^{-2}$

17 $N(x) = (6x - 7)^3(8x^2 + 9)^2$

18 $f(w) = (2w^2 - 3w + 1)(3w + 2)^4$

19 $g(z) = \left(z^2 - \frac{1}{z^2}\right)^6$

20 $S(t) = \left(\frac{3t + 4}{6t - 7}\right)^3$

21 $k(r) = \sqrt[3]{8r^3 + 27}$

22 $h(z) = (2z^2 - 9z + 8)^{-2/3}$

23 $F(v) = \frac{5}{\sqrt[3]{v^5 - 32}}$

25 $g(w) = \frac{w^2 - 4w + 3}{w^{3/2}}$

27 $H(x) = \frac{2x + 3}{\sqrt{4x^2 + 9}}$

29 $k(x) = \sin(x^2 + 2)$

31 $H(\theta) = \cos^5 3\theta$

33 $g(z) = \sec(2z + 1)^2$

35 $H(s) = \cot(s^3 - 2s)$

37 $f(x) = \cos(3x^2) + \cos^2 3x$

39 $F(\phi) = \csc^2 2\phi$

41 $K(z) = z^2 \cot 5z$

43 $h(\theta) = \tan^2 \theta \sec^3 \theta$

45 $N(x) = (\sin 5x - \cos 5x)^5$

47 $T(w) = \cot^3(3w + 1)$

24 $k(s) = \frac{1}{\sqrt{3s - 4}}$

26 $K(x) = \sqrt{4x^2 + 2x + 3}$

28 $f(x) = (7x + \sqrt{x^2 + 3})^6$

30 $f(t) = \cos(4 - 3t)$

32 $g(x) = \sin^4(x^3)$

34 $k(z) = \csc(z^2 + 4)$

36 $f(x) = \tan(2x^2 + 3)$

38 $g(w) = \tan^3 6w$

40 $M(x) = \sec(1/x^2)$

42 $G(s) = s \csc(s^2)$

44 $H(u) = u^2 \sec^3 4u$

46 $p(v) = \sin 4v \csc 4v$

48 $g(r) = \sin(2r + 3)^4$

$$49 \quad h(w) = \frac{\cos 4w}{1 - \sin 4w}$$

$$50 \quad f(x) = \frac{\sec 2x}{1 + \tan 2x}$$

$$51 \quad f(x) = \tan^3 2x - \sec^3 2x$$

$$52 \quad h(\phi) = (\tan 2\phi - \sec 2\phi)^3$$

$$53 \quad f(x) = \sin \sqrt{x} + \sqrt{\sin x}$$

$$54 \quad f(x) = \tan \sqrt[3]{5 - 6x}$$

$$55 \quad k(\theta) = \cos^2 \sqrt{3 - 8\theta}$$

$$56 \quad r(t) = \sqrt{\sin 2t - \cos 2t}$$

$$57 \quad g(x) = \sqrt{x^2 + 1} \tan \sqrt{x^2 + 1}$$

$$58 \quad h(\phi) = \frac{\cot 4\phi}{\sqrt{\phi^2 + 4}}$$

$$59 \quad M(x) = \sec \sqrt{4x + 1}$$

$$60 \quad F(s) = \sqrt{\csc 2s}$$

$$61 \quad h(x) = \sqrt{4 + \csc^2 3x}$$

$$62 \quad f(t) = \sin^2 2t \sqrt{\cos 2t}$$

Exer. 63–68: (a) Find equations of the tangent line and the normal line to the graph of the equation at P . (b) Find the x -coordinates on the graph at which the tangent line is horizontal.

$$63 \quad y = (4x^2 - 8x + 3)^4; \quad P(2, 81)$$

$$64 \quad y = (2x - 1)^{10}; \quad P(1, 1)$$

$$65 \quad y = \left(x + \frac{1}{x}\right)^5; \quad P(1, 32)$$

$$66 \quad y = \sqrt{2x^2 + 1}; \quad P(-1, \sqrt{3})$$

$$67 \quad y = 3x + \sin 3x; \quad P(0, 0)$$

$$68 \quad y = x + \cos 2x; \quad P(0, 1)$$

Exer. 69–74: Find the first and second derivatives.

$$69 \quad g(z) = \sqrt{3z + 1}$$

$$70 \quad k(s) = (s^2 + 4)^{2/3}$$

$$71 \quad k(r) = (4r + 7)^5$$

$$72 \quad f(x) = \sqrt[3]{10x + 7}$$

$$73 \quad f(x) = \sin^3 x$$

$$74 \quad G(t) = \sec^2 4t$$

Exer. 75–76: Use differentials to approximate the value.

$$75 \quad \sqrt[3]{65} \quad (\text{Hint: Let } y = \sqrt[3]{x}.)$$

$$76 \quad \sqrt[3]{35}$$

77 If an object of mass m has velocity v , then its kinetic energy K is given by $K = \frac{1}{2}mv^2$. If v is a function of time t , use the chain rule to find a formula for dK/dt .

78 As a spherical weather balloon is being inflated, its radius r is a function of time t . If V is the volume of the balloon, use the chain rule to find a formula for dV/dt .

79 When a space shuttle is launched into space, an astronaut's body weight decreases until a state of weightlessness is achieved. The weight W of a 150-pound astronaut at an altitude of x kilometers above sea level is given by

$$W = 150 \left(\frac{6400}{6400 + x} \right)^2.$$

If the space shuttle is moving away from the earth's surface at the rate of 6 km/sec, at what rate is W decreasing when $x = 1000$ km?

80 The length-weight relationship for Pacific halibut is well described by the formula $W = 10.375L^3$, where L is the length in meters and W is the weight in kilograms. The rate of growth in length dL/dt is given by $0.18(2 - L)$, where t is time in years.

(a) Find a formula for the rate of growth in weight dW/dt in terms of L .

(b) Use the formula in part (a) to estimate the rate of growth in weight of a halibut weighing 20 kilograms.

81 If $k(x) = f(g(x))$ and if $f(2) = -4$, $g(2) = 2$, $f'(2) = 3$, and $g'(2) = 5$, find $k(2)$ and $k'(2)$.

82 Let p , q , and r be functions such that $p(z) = q(r(z))$. If $r(3) = 3$, $q(3) = -2$, $r'(3) = 4$, and $q'(3) = 6$, find $p(3)$ and $p'(3)$.

83 If $f(t) = g(h(t))$ and if $f(4) = 3$, $g(4) = 3$, $h(4) = 4$, $f'(4) = 2$, and $g'(4) = -5$, find $h'(4)$.

84 If $u(x) = v(w(x))$ and if $v(0) = -1$, $w(0) = 0$, $u(0) = -1$, $v'(0) = -3$, and $u'(0) = 2$, find $w'(0)$.

c 85 Let $h = f \circ g$ be a differentiable function. The following tables list some values of f and g . Use Exercise 51 of Section 3.2 to approximate $h'(1.12)$.

x	2.2210	2.2320	2.2430
$f(x)$	4.9328	4.9818	5.0310

x	1.1100	1.1200	1.1300
$g(x)$	2.2210	2.2320	2.2430

c 86 Let $h = f \circ g$ be a differentiable function. The following tables list some values of f and g . Use Exercise 51 of Section 3.2 to approximate $h'(-2)$.

x	-8.48092	-8.46000	-8.43908
$f(x)$	-2.03930	-2.03762	-2.03594

x	-2.00400	-2.00000	-1.99600
$g(x)$	-8.48092	-8.46000	-8.43908

87 Let f be differentiable. Use the chain rule to prove that

(a) if f is even, then f' is odd

(b) if f is odd, then f' is even

Use polynomial functions to give examples of (a) and (b).

- 88 Use the chain rule, the derivative formula for $D_x \sin u$, together with the identities

$$\cos x = \sin\left(\frac{\pi}{2} - x\right) \quad \text{and} \quad \sin x = \cos\left(\frac{\pi}{2} - x\right)$$

to obtain the formula for $D_x \cos x$.

- 89 Pinnipeds are a suborder of aquatic carnivorous mammals, such as seals and walruses, whose limbs are modified into flippers. The length-weight relationship during fetal growth is well described by the formula $W = (6 \times 10^{-5})L^{2.74}$, where L is the length in centimeters and W is the weight in kilograms.

- (a) Use the chain rule to find a formula for the rate of growth in weight with respect to time t .
 (b) If the weight of a seal is 0.5 kilogram and is changing at a rate of 0.4 kilogram per month, how fast is the length changing?

- 90 The formula for the adiabatic expansion of air is $pv^{1.4} = c$, where p is the pressure, v is the volume, and c is a constant. Find a formula for the rate of change of pressure with respect to volume.

- 91 The curved surface area S of a right circular cone having altitude h and base radius r is given by $S = \pi r \sqrt{r^2 + h^2}$. For a certain cone, $r = 6$ cm. The altitude is measured as 8 centimeters, with a maximum error in measurement of ± 0.1 centimeter.

- (a) Calculate S from the measurements and use differentials to estimate the maximum error in the calculation.

- (b) Approximate the percentage error.

- 92 The period T of a simple pendulum of length l may be calculated by means of the formula $T = 2\pi\sqrt{l/g}$, where g is a gravitational constant. Use differentials to approximate the change in l that will increase T by 1%.

3.7 IMPLICIT DIFFERENTIATION

Given the equation

$$y = 2x^2 - 3,$$

we sometimes say that y is an **explicit function** of x , since we can write

$$y = f(x) \quad \text{with} \quad f(x) = 2x^2 - 3.$$

The equation

$$4x^2 - 2y = 6$$

determines the same function f , since solving for y gives us

$$-2y = -4x^2 + 6, \quad \text{or} \quad y = 2x^2 - 3.$$

For the case $4x^2 - 2y = 6$ we say that y (or f) is an **implicit function** of x , or that f is determined *implicitly* by the equation. If we substitute $f(x)$ for y in $4x^2 - 2y = 6$, we obtain

$$4x^2 - 2f(x) = 6$$

$$4x^2 - 2(2x^2 - 3) = 6$$

$$4x^2 - 4x^2 + 6 = 6.$$

The last equation is an identity, since it is true for every x in the domain of f . This is a characteristic of every function f determined implicitly by an equation in x and y ; that is, f is *implicit* if and only if substitution of $f(x)$ for y leads to an identity. Since $(x, f(x))$ is a point on the graph of f , the last statement implies that the graph of the implicit function f coincides with a portion (or all) of the graph of the equation.

In the next example we show that an equation in x and y may determine more than one implicit function.

EXAMPLE 1 How many different functions are determined implicitly by the equation $x^2 + y^2 = 1$?